CLOUD DEPLOYMENT MODELS

Assignment / Case Study

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Subject: Cloud Computing
Topics: Public, Private, Hybrid and Community Cloud

This assignment explains how deployment models can be selected for applications, compares their advantages and disadvantages, and reviews WhatsApp's infrastructure at a high level.

Task 1: Select a Deployment Model for Two Apps

App 1: Instagram — Public Cloud (best fit)

Reason 1: Instagram serves a very large audience across many countries. Public-cloud-style infrastructure can scale resources as demand changes and can serve users from multiple regions.

Reason 2: Shared cloud infrastructure and managed services can reduce the need for a company to build every server facility itself. This is a conceptual recommendation; Meta's actual infrastructure is not simply a generic third-party public cloud.

App 2: Paytm — Hybrid Cloud (best fit)

Reason 1: Payment-related and regulated workloads may need stricter control over sensitive data, access and compliance, which can be placed in a tightly controlled private environment where required.

Reason 2: Public-cloud resources can support customer-facing apps, notifications, analytics and demand spikes. A hybrid design combines control for sensitive workloads with flexible scaling for other workloads.

These are proposed choices from a CTO's perspective, not claims about the exact infrastructure used by Instagram or Paytm.

Task 2: Pros and Cons of Cloud Deployment Models

Model	Pros (at least 3)	Cons (at least 3)
Public Cloud	1. Quick to deploy. 2. Easily scales up/down. 3. Usually lower upfront hardware cost. 4. Global regions and managed services.	1. Less direct control over physical hardware. 2. Ongoing costs can grow unexpectedly. 3. Shared-responsibility security configuration is important. 4. Possible provider dependence.
Private Cloud	1. Greater control over environment. 2. Can tailor security and policies. 3. Useful for strict governance or specialized workloads. 4. Dedicated resources can offer predictable control.	1. High setup and maintenance cost. 2. Needs skilled IT staff. 3. Scaling can take longer and require extra capacity. 4. Organization carries more operational responsibility.
Hybrid Cloud	1. Sensitive work can stay in controlled systems. 2. Flexible scaling using public cloud. 3. Can support gradual migration. 4. Helps place workloads where they fit best.	1. More complex to design and manage. 2. Connecting environments securely takes work. 3. Data movement and integration can be difficult. 4. Costs and monitoring can be harder to track.
Community Cloud	1. Costs can be shared by similar organizations. 2. Policies can fit common industry needs. 3. Encourages collaboration and shared services. 4. May provide more tailored governance than a general public cloud.	1. Governance decisions can be slow. 2. Costs and responsibilities must be divided fairly. 3. Less flexibility than a fully independent setup. 4. Shared requirements may not suit every member.

Task 3: WhatsApp Cloud Infrastructure Case Study

Summary and deployment-model classification

WhatsApp is operated by Meta and serves users around the world. Public information from Meta describes a large, globally distributed infrastructure and security systems designed to support WhatsApp at scale. However, Meta does not publish a simple, complete architecture diagram that labels all of WhatsApp's production infrastructure as one textbook deployment model.

For a classroom classification, WhatsApp is best described as using a **privately operated global infrastructure (closest to a private-cloud model)**, rather than assuming it runs entirely on a third-party public cloud. This is a reasoned high-level classification; the precise mix of infrastructure technologies and any workload-specific arrangements are not fully public.

Why this approach suits WhatsApp:

• **Scalability:** A globally used messaging service must handle changing traffic and very large numbers of messages, calls and media transfers. Distributed infrastructure can add capacity and spread workloads.

• **Security and privacy:** WhatsApp uses end-to-end encryption for personal messages and calls. Meta also describes additional safeguards such as key transparency and confidential computing for specific features. Encryption is a security design, not itself a cloud deployment model.

• **Global reach and resilience:** Infrastructure distributed across locations can help serve users worldwide and improve resilience when one component or location has a problem.

• **Operational control:** Operating infrastructure directly gives Meta substantial control over hardware, software and security design, while requiring significant engineering investment.

Important distinction: WhatsApp's user-facing service is available publicly, but that does not automatically mean its underlying infrastructure is the “Public Cloud” deployment model. Public availability and cloud deployment model are different concepts.

Task 4: IPL Ticket Booking Platform

Chosen model: Public Cloud

I would choose a public cloud for a new online IPL ticket-booking platform because traffic can rise sharply when tickets go on sale, and public cloud services provide flexible capacity without purchasing all physical servers in advance.

Two possible risks:

1. **Sudden traffic spikes:** A large number of fans may visit at the same time, causing slow responses or outages.

2. **Security or data exposure:** Incorrect access settings or application vulnerabilities could expose customer information or disrupt bookings.

Mitigation for Risk 1: Use autoscaling, a load balancer, a content delivery network (CDN), caching, and load testing before ticket release. Also place users in a controlled virtual waiting room if needed, and monitor capacity during sales.

Additional good practice: Use multi-factor authentication for administrators, least-privilege access, encryption, backups, monitoring and a tested incident-response plan.

References (online)

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• Meta, “Meta Partners With Reliance on AI-Enabled Data Center in India” (2026): https://about.fb.com/news/2026/06/meta-partners-with-reliance-on-ai-enabled-data-center-in-india/